

Muhammad Daniyal Nadeem

Islamabad, Pakistan

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🔗 Daniyal-engineer (Portfolio)

Professional Summary

Computer Engineering student at Air University with strong knowledge of Programming, Data Structures, Object-Oriented Programming, Embedded Systems, Digital Signal Processing, Artificial Intelligence, Robotics, and IoT systems. Completed six semesters of the Bachelor of Computer Engineering program and worked on software, machine learning, and robotics projects.

Education

Air University

Bachelor of Computer Engineering (PEC Accredited)
Completed 6 Semesters

Islamabad

2023–2027

Experience

OGDCL – System Department

Intern

Islamabad

Summer 2025

- Worked within OGDCL's system department, supporting system operations and maintenance tasks.
- Gained hands-on exposure to Microsoft Azure cloud services and enterprise IT infrastructure.
- Assisted the departmental team in monitoring and troubleshooting system-related issues.

Punjab Auto Workshop

Intern

Islamabad

July 2026

- Completed technical training and assisted with various workshop operations and tasks.

Projects

MUHAFIZ - Autonomous Visitor Guidance and Waste Awareness Robot

Final Year Project

- Designing an autonomous mobile robot featuring a 40x40 cm chassis, 4 wheels, a 7-inch touchscreen, and a camera module.
- Integrated an NVIDIA Jetson Nano 4GB, ESP32, and BTS7960 motor drivers.
- Implemented a power management system using buck converters and a 12V battery system.

Traffic Sign Detection System

DSP Project

- Developed a machine learning model for traffic sign recognition and classification.
- Applied image preprocessing, feature extraction, and model training techniques.
- Evaluated model performance using real-world traffic sign datasets.

Maze Solver Robot

Microprocessor Project

- Designed and implemented an autonomous maze-solving robot using sensors and microcontroller programming.
- Developed navigation logic for obstacle detection and path selection.
- Integrated sensor feedback with motor control for autonomous movement.

Car Rental System

OOP Project

- Developed a complete car rental management system in C++.
- Implemented inheritance, polymorphism, encapsulation, and abstraction.
- Managed customer, vehicle, and booking operations.

Bank Management System

DSA Project

- Implemented linked lists, stacks, sorting, and searching algorithms.
- Developed account management and transaction tracking functionalities.
- Improved data retrieval and customer account handling efficiency.

Technical Skills

Languages: C, C++, Python

Core Areas: OOP, Data Structures & Algorithms, Problem Solving

AI/ML: Machine Learning Basics, Traffic Sign Detection

Embedded Systems: Arduino, Sensor Integration, Microprocessors

Tools: MATLAB, Proteus, VS Code, Git

Databases: MS SQL Server

Interests & Research Areas

- **Robotics & Hardware Development:** Autonomous mobile systems, obstacle detection, and offline voice recognition hardware.
- **Digital Signal & Image Processing:** Spatial vision algorithms, structured light 3D tracking, and frequency domain filtering.
- **Embedded Systems & Logic Design:** Microcontrollers, single-board computers, and Verilog logic design in Xilinx ISE.
- **Finance & Investment:** Active interest in stock market investing and trading on the Pakistan Stock Exchange (PSX).
- **Career Interests:** Exploring software engineering, DevOps, and robotics R&D.